

Assessment Tool CO3-AT-1

SIMATS ENGINEERING

Course Name: Computer vision

Course Code: ITA0508

I N. Harshith / 192325078 certify that the submission is my original work and that this submission is my original work and that I have adhered to the guidelines integrity rules will result in disciplinary action.

85%.

Criteria.	weightage	Level 4	Level 3	Level 2	level 1
Understanding of concepts	25 20✓	Demonstrates through Understanding	Good. Understanding with.	Basic Understanding	Poor. understanding
Experimental design	30. 25✓	well-planned. systematic	Correct Implementation	Basic Implementation	Incorrect or incomplete
Use of tools.	20. 15✓	Effective and. advanced.	Appropriate use of tools.	Limited or. inconsistent	Incorrect or minimal.
Analysis of Results.	15 13	Insightful analysis.	Good. analysis.	Basic analysis.	Poor.
Documentation.	10 8✓	clear well structured.	organized documentation	Basic documentation.	Poor

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Experiments

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

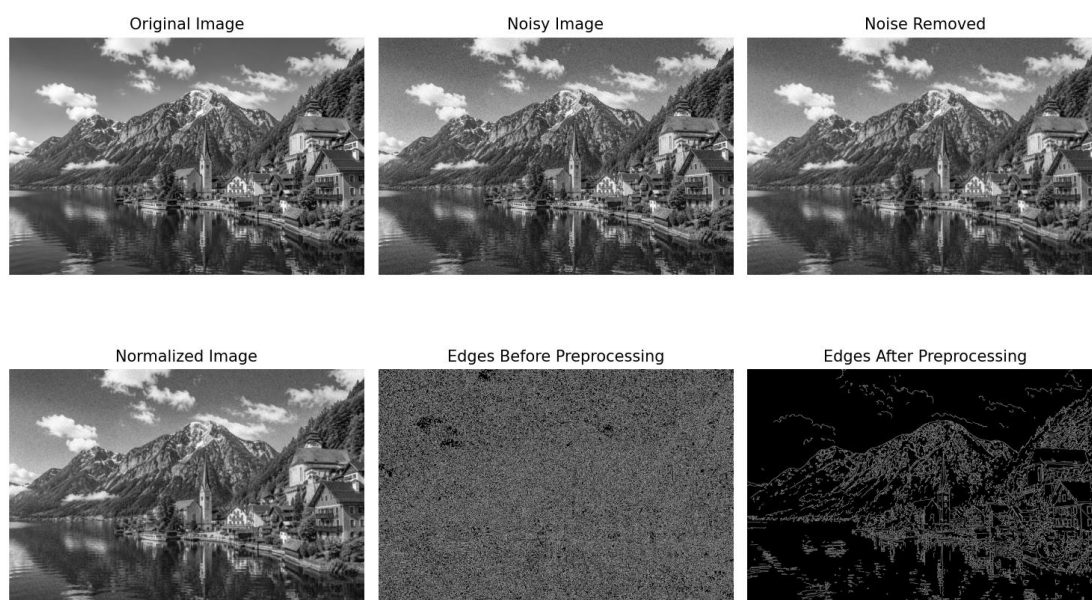
CO3: *Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BTL 4)*

Assessment Tool Weightage: 40%

Instructions: Perform the experiment using the laboratory performance rubric Understanding of Concepts, Experimental Design, Use of Tools, Analysis of Results, Documentation.

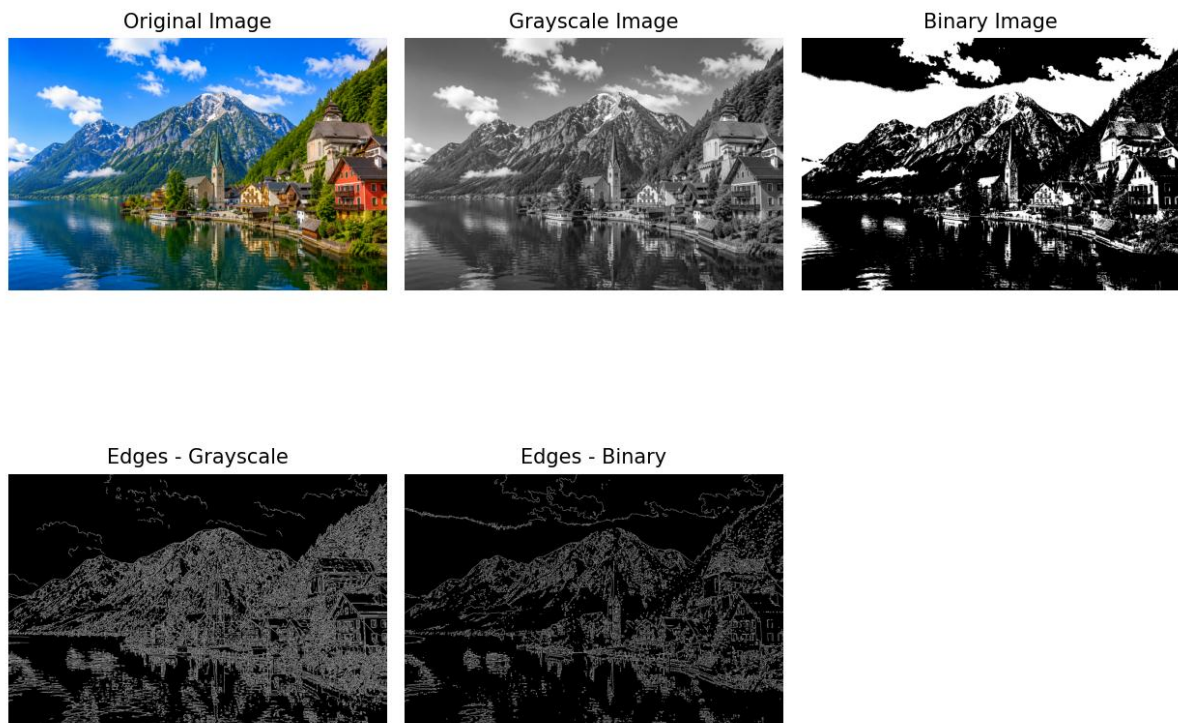
1. Preprocessing Impact Analysis

Design and perform an experiment to study the impact of preprocessing techniques such as noise removal and normalization on image quality and feature detection. Explain the theoretical concepts behind preprocessing methods and justify the selection of techniques used. Implement the preprocessing process using suitable software/tools, document the workflow and outputs systematically, and analyze how preprocessing influences feature detection accuracy and reliability.



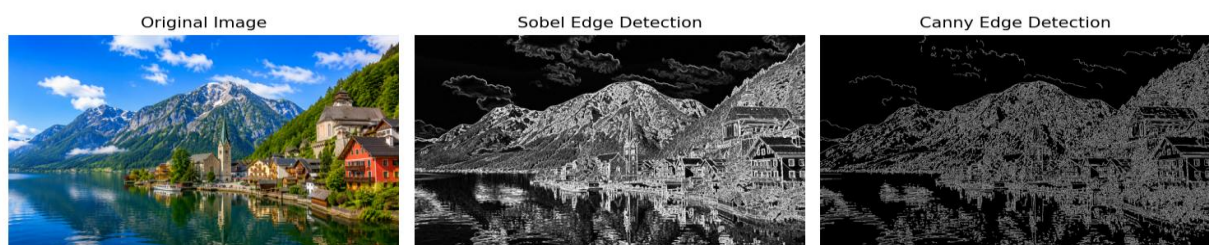
2. Image Representation Study

Conduct an experiment to convert a given image into grayscale and binary representations. Explain the significance of different image representations in computer vision applications. Design the conversion procedure using appropriate tools, document the generated outputs clearly, and analyze how image representation affects edge detection and feature extraction performance.



3. Edge Detection Comparison

Design an experiment to compare Sobel and Canny edge detection techniques on the same input image. Explain the underlying concepts and operational principles of both methods. Apply the techniques using suitable tools/software, document the detected edges systematically, and analyze the comparative effectiveness of each method for different image conditions and applications.



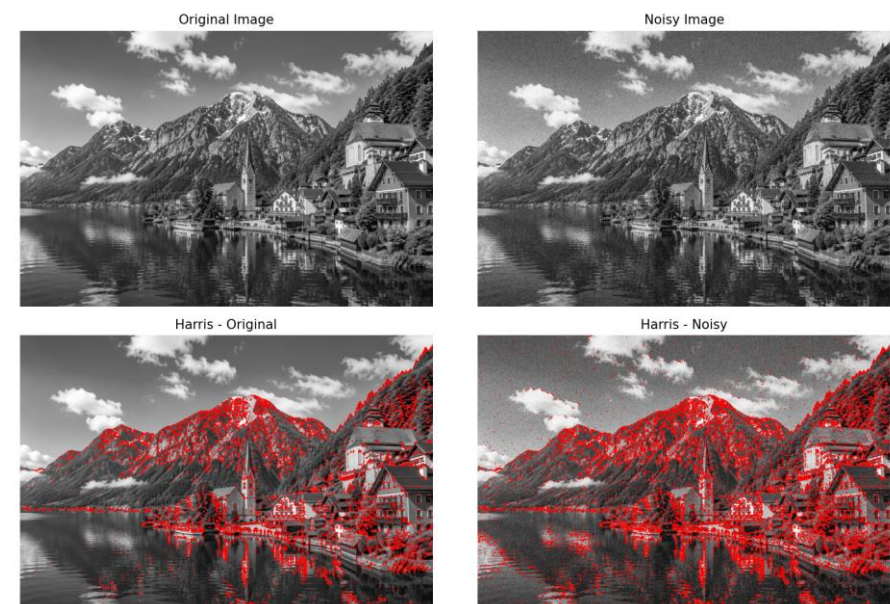
4. Harris Corner Detection Analysis

Perform Harris corner detection on images containing varying textures and patterns. Explain the concept of corner detection and its importance in computer vision tasks. Use appropriate parameters and tools to execute the experiment, document the detected corner points clearly, and analyze the distribution, accuracy, and significance of the detected corners.



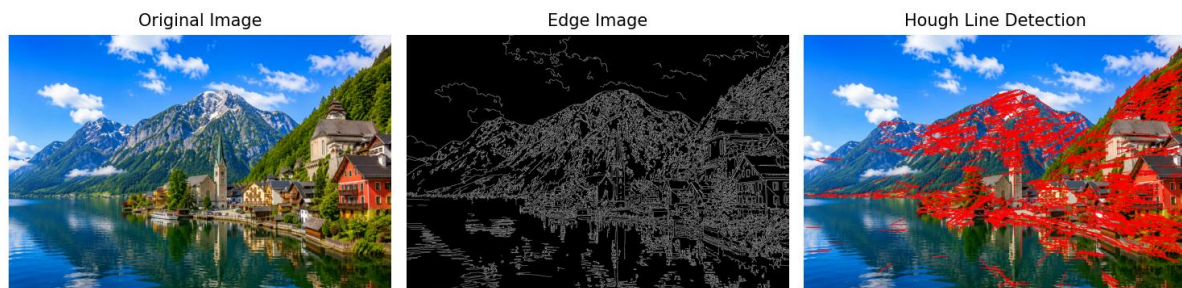
5. Effect of Noise on Corner Detection

Design an experiment to study the effect of noise on Harris corner detection performance. Explain the purpose of introducing noise and its influence on image features. Add noise systematically, perform corner detection using suitable tools, document observations for original and noisy images, and analyze the impact of noise along with possible improvement techniques.



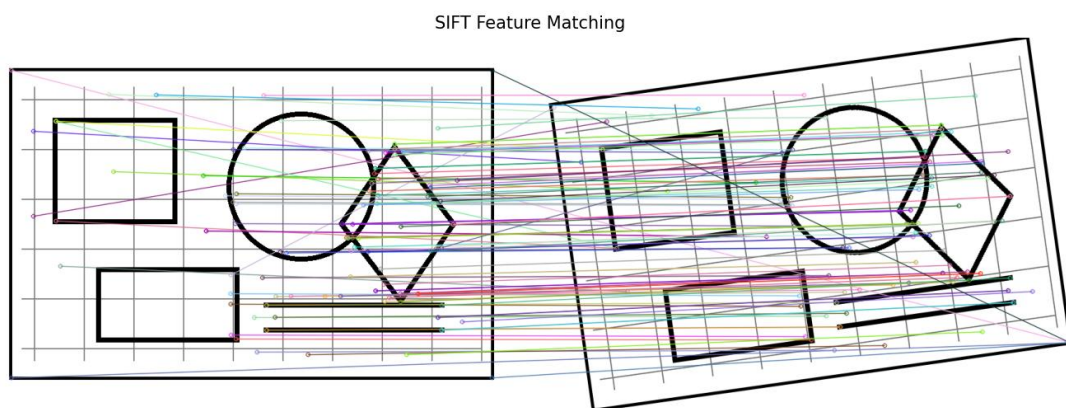
6. Hough Transform for Shape Detection

Conduct an experiment using the Hough Transform to detect lines or circles in an image. Explain the principles and significance of the Hough Transform in shape detection. Execute the detection process using appropriate tools, record the detected shapes and their parameters, and analyze the detection accuracy under different image conditions.



7. Feature Matching using SIFT

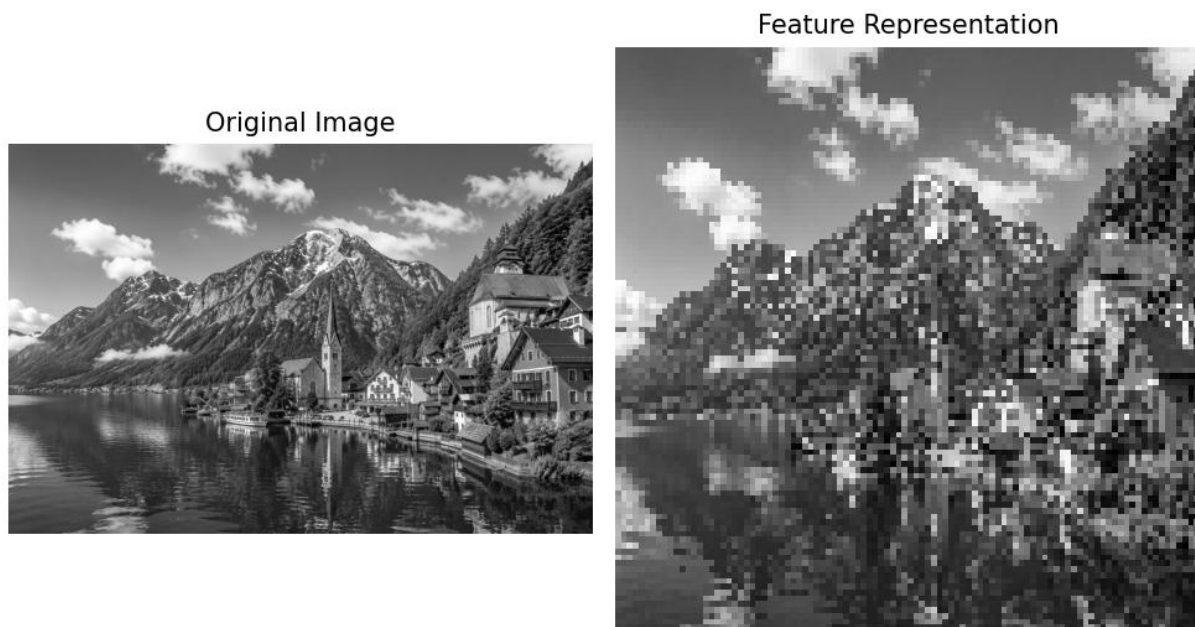
Perform feature extraction and matching between two images using the SIFT algorithm. Explain the concept of feature matching and scale-invariant features. Use suitable software/tools to extract and match features, document the matching key points systematically, and analyze the robustness of SIFT under variations in scale, rotation, and illumination.



8. PCA for Feature Reduction

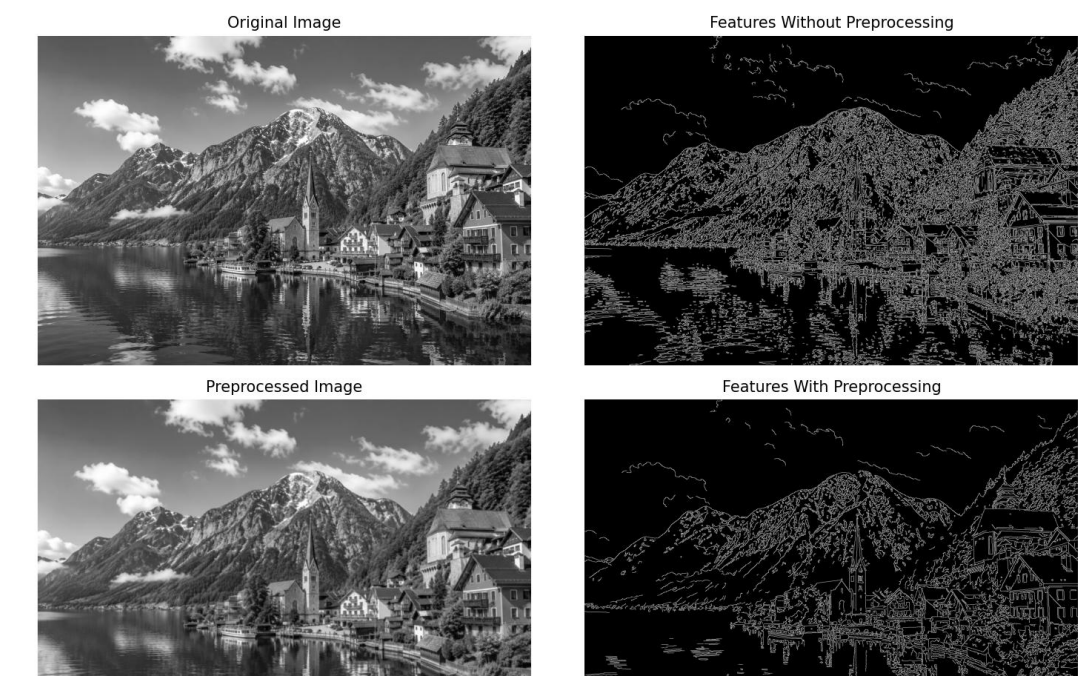
Design and implement an experiment using Principal Component Analysis (PCA) for dimensionality reduction of extracted image features. Explain the importance of feature reduction in machine learning and computer vision applications. Apply PCA

using appropriate tools, document the feature dimensions before and after reduction, and analyze its impact on computational efficiency and feature representation quality.



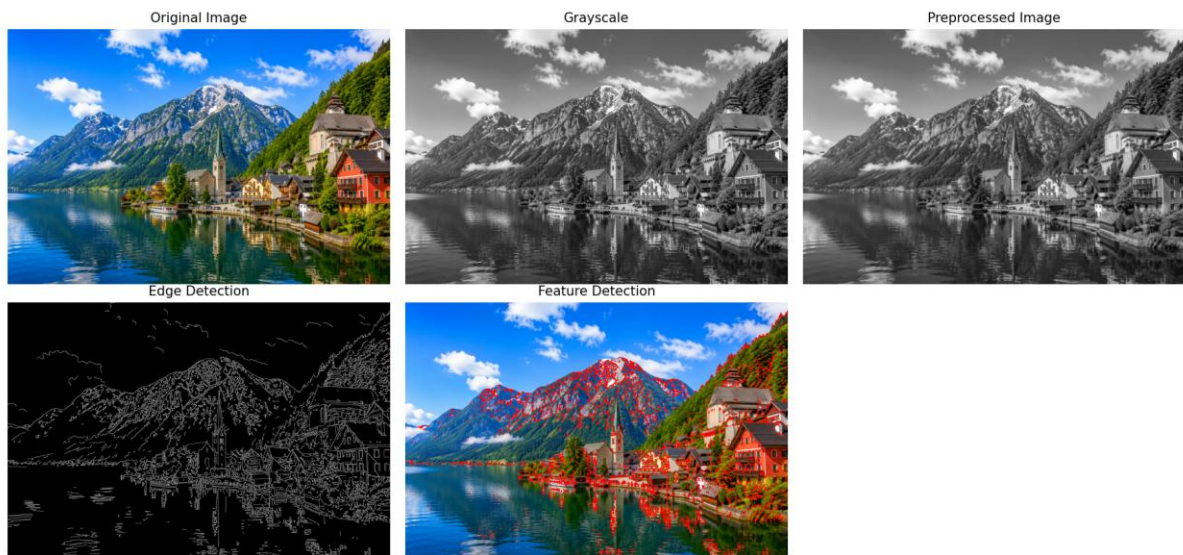
9. Preprocessing vs Feature Extraction

Conduct a comparative study of feature extraction performed with and without preprocessing. Explain the relationship between preprocessing and feature extraction performance. Execute both approaches using suitable tools, document the observations systematically, and analyze the differences in feature quality, detection accuracy, and overall system performance.



10. Integrated Feature Detection Pipeline

Design and implement a complete image processing pipeline consisting of preprocessing, edge detection, and feature extraction stages. Explain the purpose and workflow of the integrated pipeline. Execute each stage systematically using appropriate tools, document intermediate and final outputs clearly, and analyze the contribution and importance of each stage in achieving accurate feature detection results.



Code of Conduct:

I N.Harshith/192325078 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:**RUBRICS FOR CALCULATION:**

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates	Good understanding with proper integration	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts

		multiple concepts/technologies	of most concepts		
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools

Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, wellstructured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation